

CYBERSECURITY & ETHICAL HACKING

Professional Training Curriculum

A structured curriculum covering cybersecurity fundamentals, network and system security, application security, SOC operations, incident response, digital forensics, ethical hacking, vulnerability assessment, penetration testing, and a live practical project.

PROGRAM STRUCTURE

12 Cybersecurity Modules + 12 Ethical Hacking Modules

PART I — CYBERSECURITY

Module 1 — Cybersecurity Fundamentals

- Cybersecurity, CIA Triad & Security Principles
- Threats, Vulnerabilities, Risks & Attack Lifecycle
- Cybersecurity Frameworks & Standards
- Cyber Ethics, Laws & Compliance

Module 2 — Networking & Network Security

- Networking, TCP/IP, Protocols & Network Architecture
- Firewalls, VPN, IDS/IPS & Network Segmentation
- Wireless & Network Access Security
- Network Monitoring & Traffic Analysis

Module 3 — System & Endpoint Security

- Windows & Linux Security
- Users, Permissions & Access Control
- System Hardening & Patch Management
- Endpoint Security, Antivirus & EDR

Module 4 — Identity & Access Management

- Authentication, Authorization & MFA
- Password & Credential Security
- Privileged Access Management
- Identity & Access Controls

Module 5 — Cryptography & Data Security

- Encryption, Hashing & Digital Signatures
- Symmetric & Asymmetric Cryptography
- SSL/TLS, PKI & Certificates
- Data Protection & Secure Storage

Module 6 — Application & Web Security

- Web & Application Security Fundamentals
- OWASP Top 10 & Secure Development
- Authentication, Session & Access Control
- API Security & Secure Coding

Module 7 — Cloud & IoT Security

- Cloud Security & IAM
- Cloud Threats & Misconfigurations
- IoT Architecture, Threats & Security
- Security Controls for Cloud & IoT

Module 8 — Malware & Social Engineering

- Malware, Ransomware & Attack Techniques
- Phishing, Social Engineering & Human Risk
- Malware Prevention & Endpoint Protection
- Security Awareness & Response

Module 9 — SOC & Security Monitoring

- SOC Fundamentals & Security Operations
- Log Analysis & SIEM
- Threat Intelligence & IOC Analysis
- Security Alerts & Incident Detection

Module 10 — Incident Response

- Incident Response Lifecycle
- Detection, Analysis & Containment
- Eradication, Recovery & Reporting
- Incident Response Practical Exercises

Module 11 — Digital Forensics & Risk Management

- Digital Forensics Fundamentals
- Disk, Network & Log Forensics
- Risk Assessment & Vulnerability Management
- Security Audit, Policies & Compliance

Module 12 — Live Cybersecurity Project

- Security Assessment of a Simulated Organization
- SOC Monitoring & Incident Investigation
- Vulnerability Identification & Remediation
- Final Security Report & Presentation

PART II — ETHICAL HACKING

Module 1 — Ethical Hacking Fundamentals

- Ethical Hacking & Penetration Testing
- Hacker Types, Attack Lifecycle & Methodologies
- Cyber Kill Chain & MITRE ATT&CK;
- Scope, Authorization, Laws & Ethics

Module 2 — Footprinting & Reconnaissance

- Passive & Active Reconnaissance
- OSINT, WHOIS & DNS
- Search Engines, Social Media & Email Footprinting
- Subdomain & Attack Surface Discovery

Module 3 — Scanning & Enumeration

- Network & Host Discovery
- Port, Service & OS Scanning
- Nmap & Network Scanning Techniques
- SMB, DNS, SNMP & LDAP Enumeration

Module 4 — Vulnerability Assessment

- Vulnerability Identification & Classification
- CVE, CVSS & Vulnerability Research
- Automated Vulnerability Scanning
- Vulnerability Validation & Reporting

Module 5 — System Hacking

- Password Attacks & Hash Cracking
- Authentication & Credential Attacks
- Privilege Escalation
- System Exploitation & Post-Exploitation

Module 6 — Network Hacking

- Packet Sniffing & Traffic Analysis
- ARP Spoofing & MITM
- Session Hijacking
- DoS/DDoS & Network Attack Testing

Module 7 — Social Engineering & Malware

- Social Engineering & Phishing
- Spear Phishing & Impersonation
- Malware, Trojans & Ransomware
- Attack Detection & Countermeasures

Module 8 — Network Defense Evasion

- Firewalls & IDS/IPS
- Honeypots & Detection Mechanisms
- Security Control Testing
- Evasion Concepts & Countermeasures

Module 9 — Web Server & Web Application Hacking

- Web Server Reconnaissance & Enumeration
- Web Application Reconnaissance & Testing
- OWASP Top 10 Vulnerabilities
- Authentication, Session & Access-Control Attacks

Module 10 — SQL Injection & API Security

- SQL Injection Fundamentals
- Authentication Bypass & Blind SQLi
- API Reconnaissance & Testing
- API Authentication & Authorization

Module 11 — Wireless, Mobile, IoT & Cloud Hacking

- Wireless Security Testing
- Mobile Application Security Testing
- IoT/OT Security Testing
- Cloud Security Assessment

Module 12 — Live Ethical Hacking Project

- Complete Reconnaissance & Vulnerability Assessment
- Controlled Exploitation & Validation
- Risk Assessment & Remediation
- Professional VAPT Report & Presentation

Note: Practical security testing and exploitation activities should be performed only in authorized, controlled environments such as labs, simulations, CTFs, or systems for which explicit permission has been obtained.